

Week 7: AI Agents

Course: BUSM2725 AI and Analytics for Business Applications

Outline

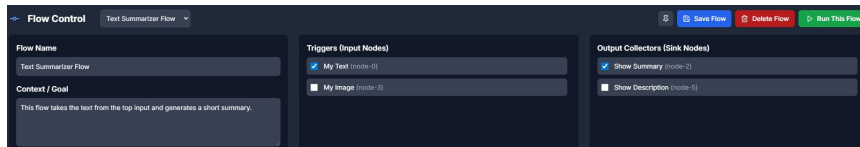
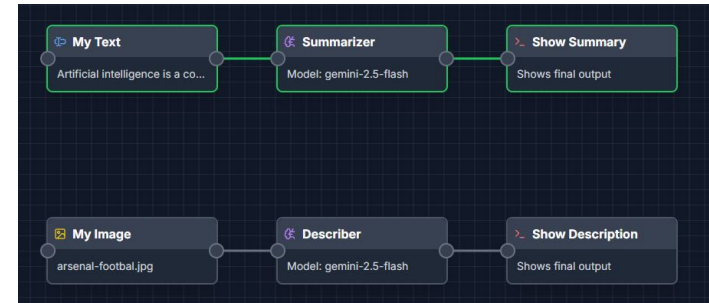
- Overview of AI agents
- Activity: Hands-on with SPARK with images, documents, and tools
- Activity: Retrieval-Augmented Generation (RAG)
- Activity: Build AI agent with SPARK
- Activity: Build AI agent for analysing manufacturing data with Gemini
- Q&A for Assessment 2

Move to MS Learn Slides

- Define what AI agents are
- Explain the business value of AI agents
- Identify common use cases for AI agents
- Describe the core components that make up AI agents

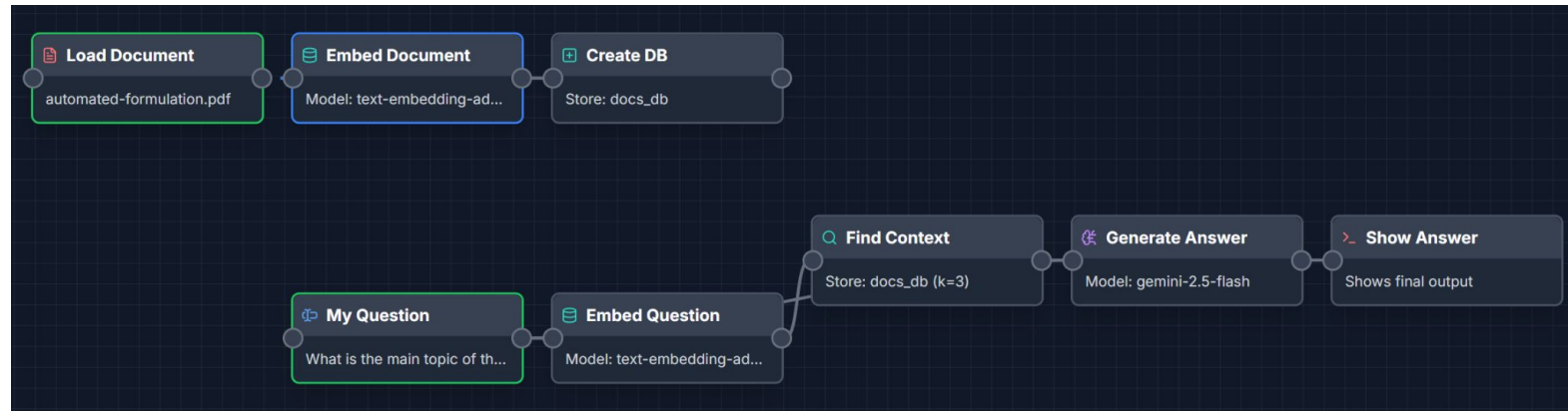
Activity: Hands-on with SPARK with images, documents, and tools

- Open SPARK app
- Open Dial Flow Test example
- You can build multiple flows on the canvas
- Move your mouse to the bottle to open Flow Control Panel
- You can define your flow (inputs/outputs) and run each flow

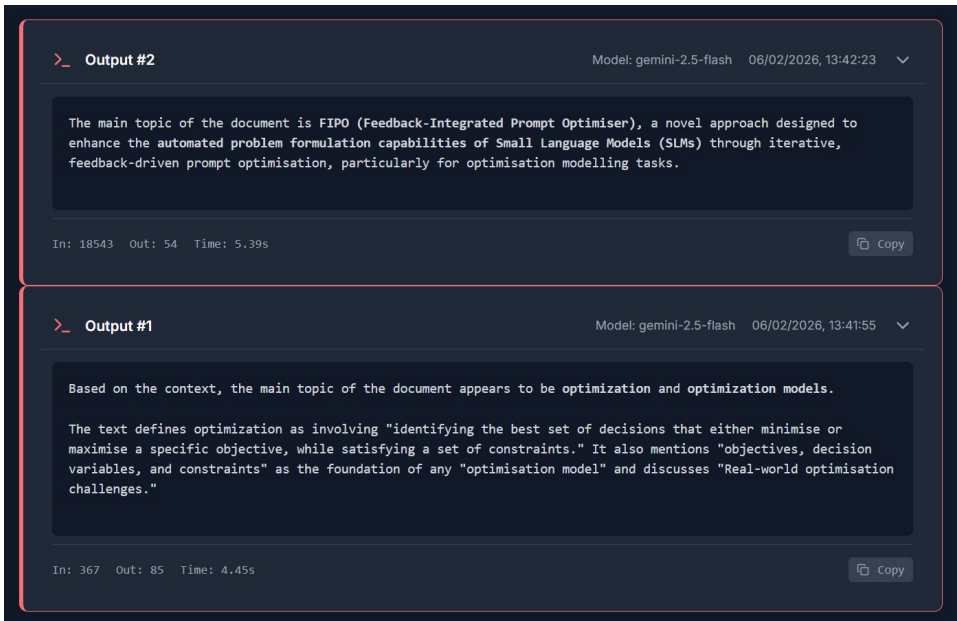


Demo: Retrieval-Augmented Generation (RAG)

- Open SPARK app
- Open the Q&A with RAG (Embedding might not work from your end because of updates in APIs)



Token Consumption



>_ Output #2 Model: gemini-2.5-flash 06/02/2026, 13:42:23

The main topic of the document is FIPO (Feedback-Integrated Prompt Optimiser), a novel approach designed to enhance the automated problem formulation capabilities of Small Language Models (SLMs) through iterative, feedback-driven prompt optimisation, particularly for optimisation modelling tasks.

In: 18543 Out: 54 Time: 5.39s Copy

>_ Output #1 Model: gemini-2.5-flash 06/02/2026, 13:41:55

Based on the context, the main topic of the document appears to be optimization and optimization models.

The text defines optimization as involving "identifying the best set of decisions that either minimise or maximise a specific objective, while satisfying a set of constraints." It also mentions "objectives, decision variables, and constraints" as the foundation of any "optimisation model" and discusses "Real-world optimisation challenges."

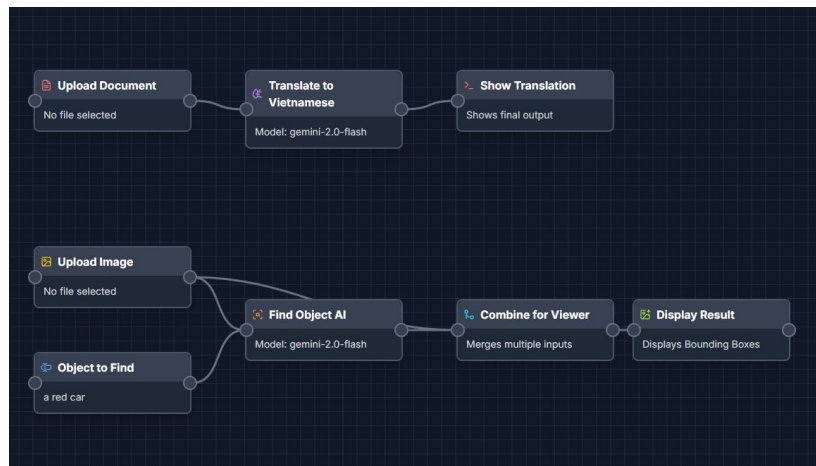
In: 367 Out: 85 Time: 4.45s Copy

Feed all file content to LLM

Use RAG

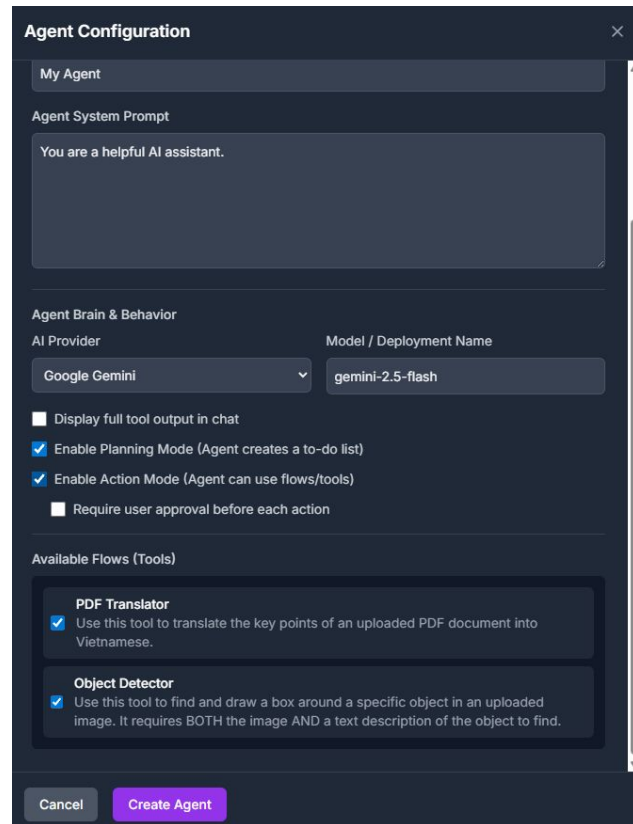
Activity: Build AI agent with SPARK.

- Open “Conversational Agent Demo” example in the Gallery
- You will see two flows
- Open Flow Control Panel to review each flow and its context
- Test each flow to see if it works as expected



Build Agent for existing flows

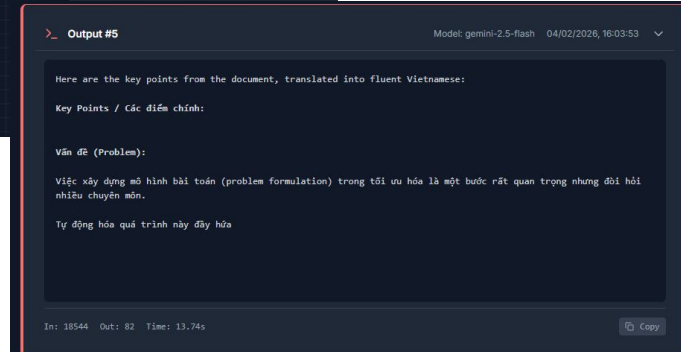
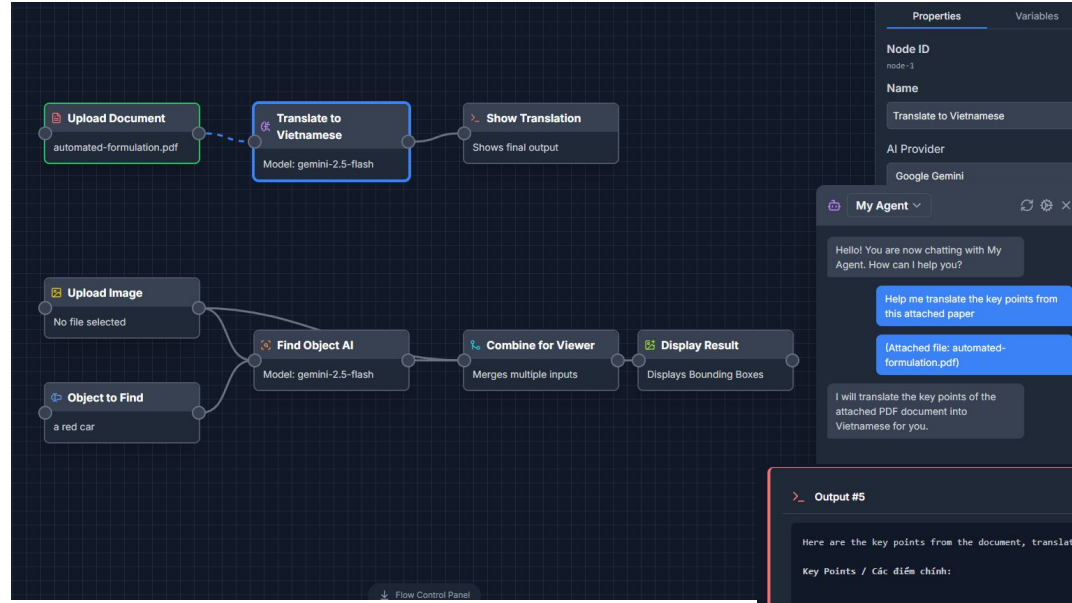
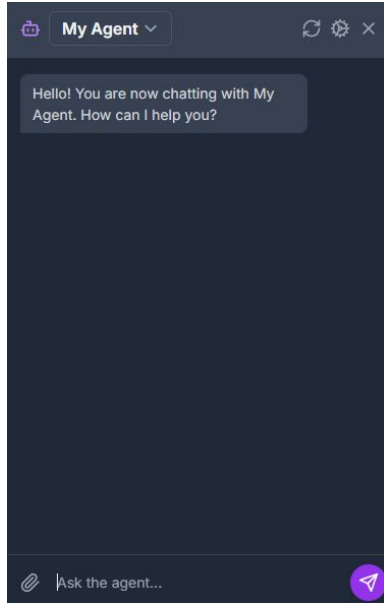
- From the previous example, click **Agentify**
- Click New Agent
- Provide name, prompt, etc.
- Check the boxes for all available Flows (Tools)
- Click Create Agent
- Test your agent (like ChatGPT or Val)



The image shows a 'Agent Configuration' dialog box with a dark theme. It contains the following sections:

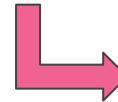
- Agent Name:** A text input field containing 'My Agent'.
- Agent System Prompt:** A text area containing 'You are a helpful AI assistant.'
- Agent Brain & Behavior:**
 - AI Provider:** A dropdown menu showing 'Google Gemini'.
 - Model / Deployment Name:** A text input field containing 'gemini-2.5-flash'.
 - Options:**
 - ☐ Display full tool output in chat
 - ☒ Enable Planning Mode (Agent creates a to-do list)
 - ☒ Enable Action Mode (Agent can use flows/tools)
 - ☐ Require user approval before each action
- Available Flows (Tools):**
 - PDF Translator:** ☒ Use this tool to translate the key points of an uploaded PDF document into Vietnamese.
 - Object Detector:** ☒ Use this tool to find and draw a box around a specific object in an uploaded image. It requires BOTH the image AND a text description of the object to find.
- Buttons:** 'Cancel' and 'Create Agent' at the bottom.

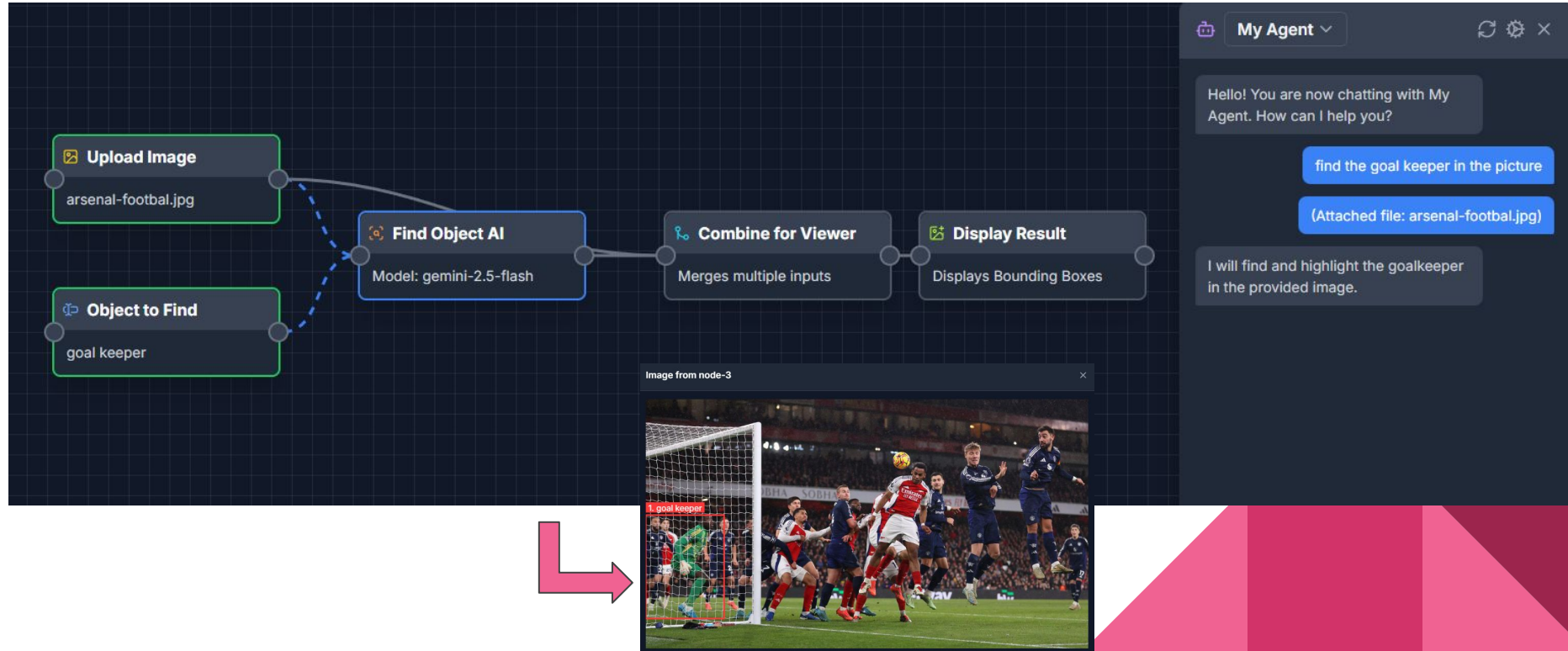
Testing Agent (Translation Flow)



- ☒ Display full tool output in chat
- ☒ Enable Planning Mode (Agent creates a to-do list)
- ☒ Enable Action Mode (Agent can use flows/tools)
- ☐ Require user approval before each action

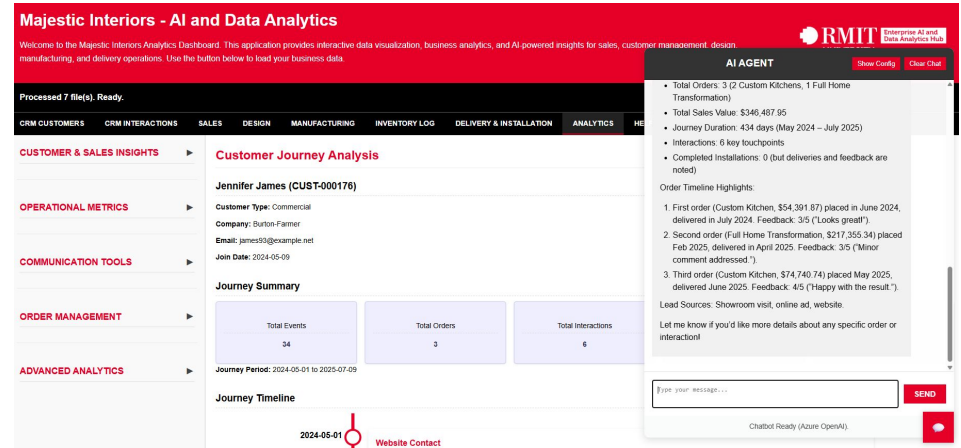
Check the Display Full Tool output to show the result right in the chat box.





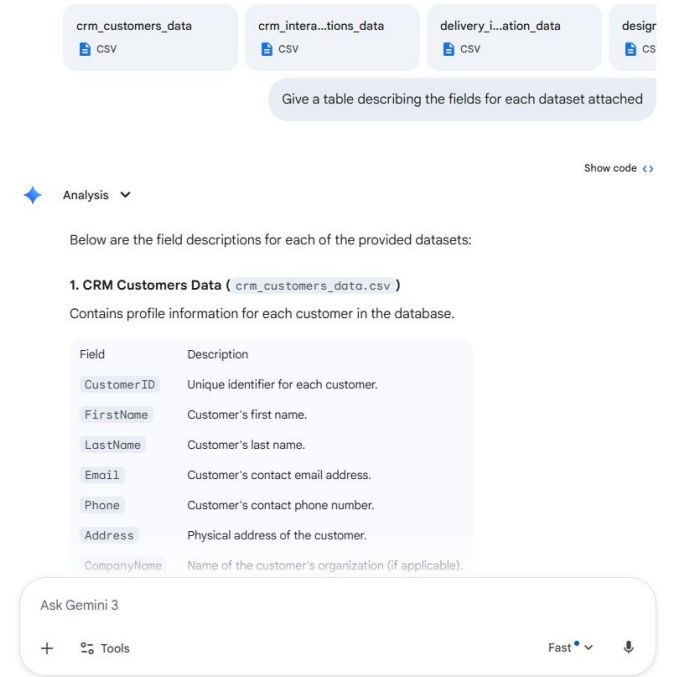
Activity: Build AI agent for analysing manufacturing data with Gemini

- Review the AI for Manufacturing (Majestic Interiors)
- Review the introduction video (on Course Canvas) to see example prompts
- Select one analytics area (e.g. sales) that you are interested in



Data Overview

- Attached the 7 csv files from this example to Gemini and ask: “Give a table describing the fields for each dataset attached”



The screenshot shows the Gemini AI interface with four dataset files attached: `crm_customers_data.csv`, `crm_interactions_data.csv`, `delivery_location_data.csv`, and `design_data.csv`. A prompt asks for a table describing the fields for each dataset. The AI response, under the 'Analysis' tab, provides field descriptions for the 'CRM Customers Data' dataset.

Below are the field descriptions for each of the provided datasets:

1. CRM Customers Data (`crm_customers_data.csv`)
Contains profile information for each customer in the database.

Field	Description
<code>CustomerID</code>	Unique identifier for each customer.
<code>FirstName</code>	Customer's first name.
<code>LastName</code>	Customer's last name.
<code>Email</code>	Customer's contact email address.
<code>Phone</code>	Customer's contact phone number.
<code>Address</code>	Physical address of the customer.
<code>CompanyName</code>	Name of the customer's organization (if applicable).

At the bottom, there is an input field with the text 'Ask Gemini 3' and a 'Fast' dropdown menu.

Build App with Gemini

Build a simple app with a dashboard. The app should allow me to upload the CRM data (the csv file above) and provide numbers of visualisations to help me understand the sales performance.



Canvas x

Fast ▾





I will now build a comprehensive Sales Dashboard application.

This app will allow you to upload the CSV files you've provided (specifically targeting the Sales, Customers, and Interactions data) to generate high-level metrics and visual insights. It includes:

- **Key Performance Indicators (KPIs):** Total Revenue, Conversion Rate, and Average Order Value.
- **Sales Funnel Visualization:** Understanding the distribution of leads across different stages.
- **Revenue Trends:** Monthly performance tracking.
- **Lead Source Analysis:** Identifying which channels are driving the most value.
- **Product Performance:** Comparing estimated values across different product interests.

 CRM Sales Dashboard
Feb 4, 4:31 PM

[Try again without Canvas](#)

I have built a dashboard that focuses on the sales funnel and

Let's write or build together

+   Canvas x

Fast  v



CRM Sales Dashboard






Code
Preview
Share
x



CRM Sales Insights

Analyze your pipeline and sales performance





No Data Available

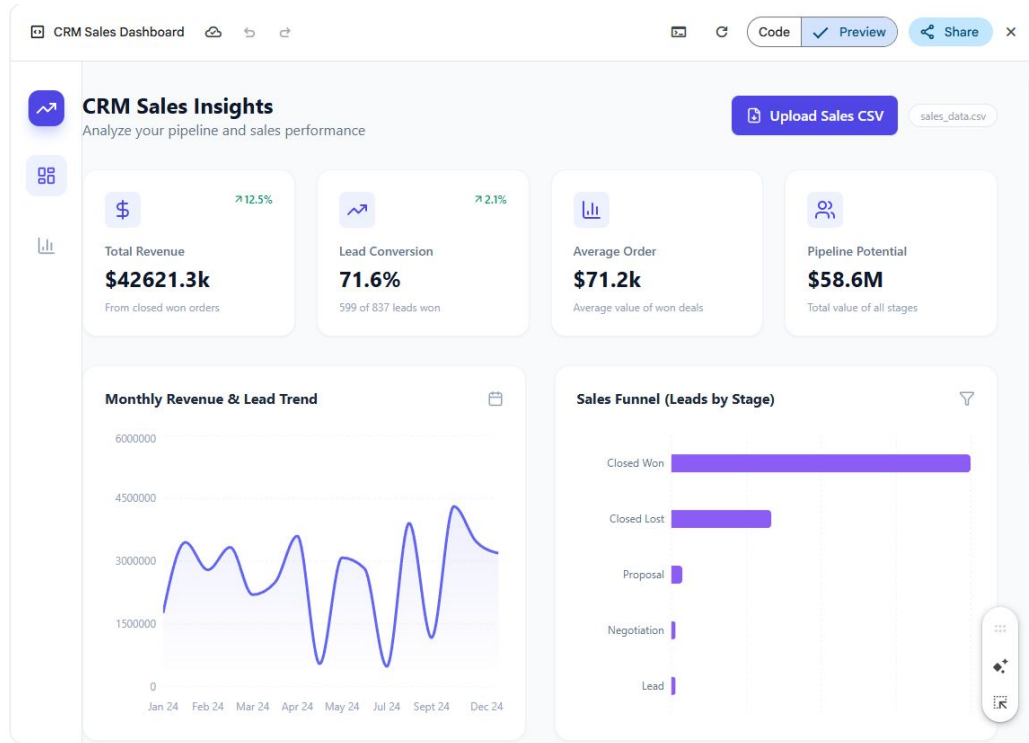
Please upload the **sales_data.csv** file to begin visualizing your business performance.

 Upload Sales CSV





Upload the related csv file and see the dashboard



You can scan through the code if you have html coding experience

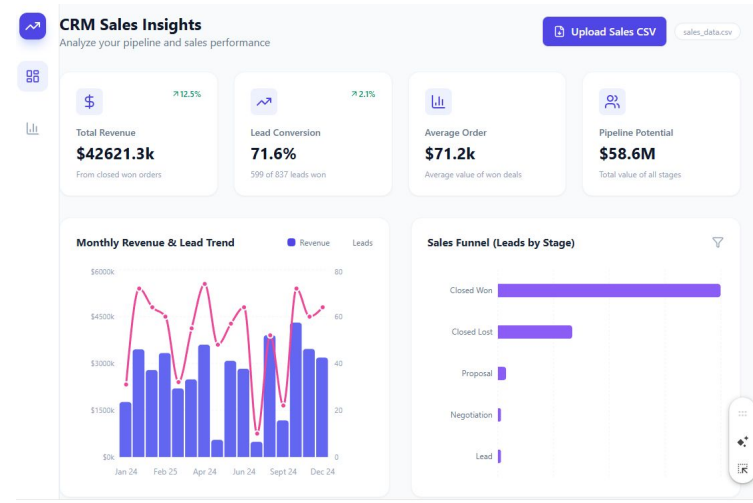
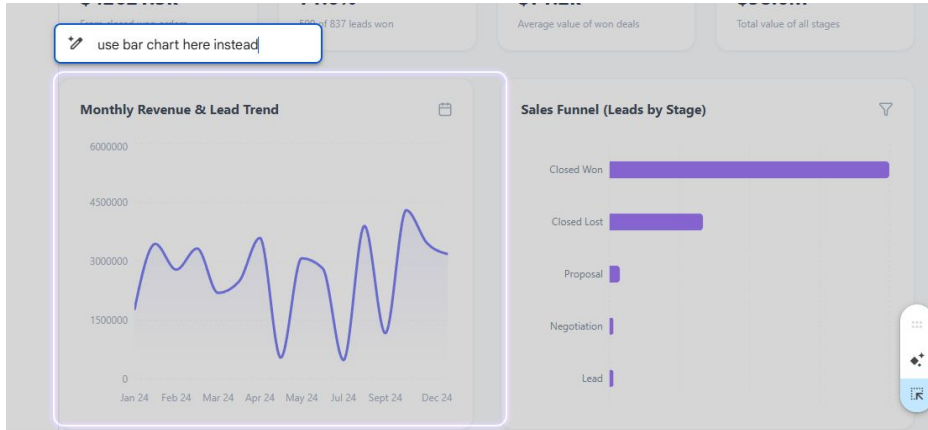
```

1 import React, { useState, useReducer, useCallback } from 'react';
2 import {
3   BarChart, Bar, XAxis, YAxis, CartesianGrid, Tooltip, Legend, ResponsiveContainer,
4   LineChart, Line, PieChart, Pie, Cell, AreaChart, Area
5 } from 'recharts';
6 import {
7   TrendingUp, Users, DollarSign, PieChart as PieIcon,
8   BarChart3, Calendar, FileDown, LayoutDashboard, Search,
9   Filter, ArrowUpRight, ArrowDownRight
10 } from 'lucide-react';
11
12 const COLORS = ['#6366f1', '#8b5cf6', '#e64999', '#f43f5e', '#f97316', '#f06a47', '#f97316', '#f06a47'];
13
14 const App = () => {
15   const [salesData, setSalesData] = useState([]);
16   const [customerData, setCustomerData] = useState([]);
17   const [fileName, setFileName] = useState('');
18   const [activeTab, setActiveTab] = useState('overview');
19
20   // Parser for CSV data
21   const handleFileUpload = (event) => {
22     const file = event.target.files[0];
23     if (file) return;
24     setFileName(file.name);
25
26     const reader = new FileReader();
27     reader.onload = (e) => {
28       const text = e.target.result;
29       const rows = text.split('\n').filter(row => row.trim() !== '');
30       const headers = rows[0].split(',').map(h => h.trim());
31
32       const parsedData = rows.slice(1).map(row => {
33         // Handle CSV comma within quotes if necessary, but keep it simple for now
34         const values = row.split(/,(?!(?:''))(?!'')/g).map(v => v.trim().replace(/"/g, ''));

```

Fine tune the app

- You can fine tune the app by providing detailed instructions
- Or you use select and ask



Add AI features

add a new feature that allow user to ask question about the data and the app will filter the data for the dashboard based on users' prompt (e.g. revenue from April to September for Full Home Transformation).

+  Canvas x

Fast 



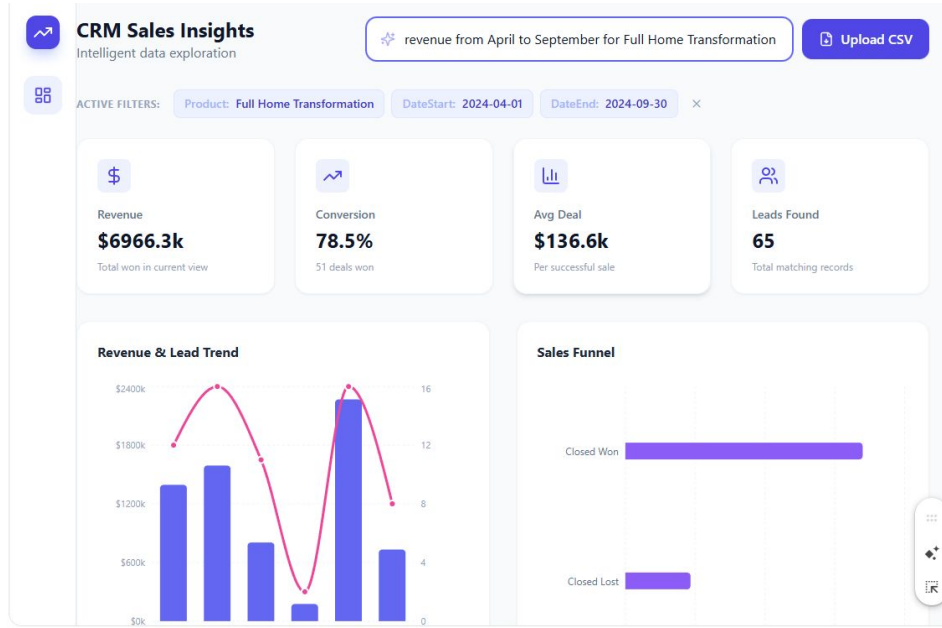
I will now integrate a natural language search feature into the dashboard.

This feature uses the Gemini 2.5 Flash model to interpret your questions and return structured filters (like date ranges, product types, or sales stages). The app will then apply these filters to the dashboard metrics and visualizations in real-time, allowing you to slice and dice the sales data just by asking.

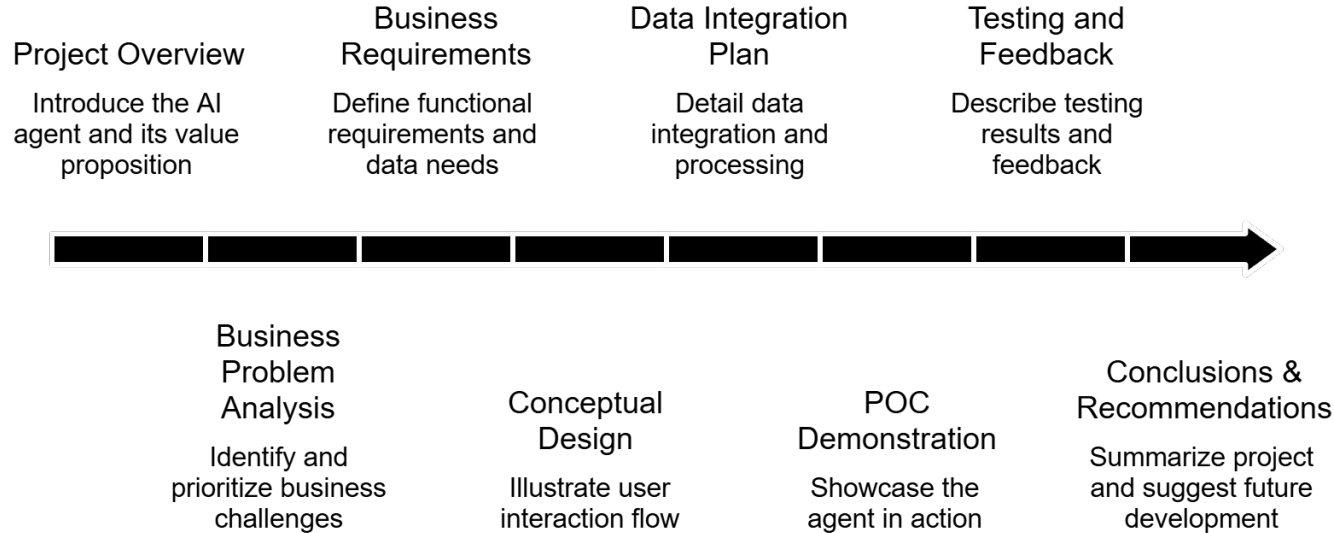
The new feature is now live!

How to use it:

1. Upload your `sales_data.csv` as before.
2. Use the "Sparkle" search bar at the top. You can type queries like:



Q&A for Assessment 2



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